

So what's wrong with complicated models?

- ❶ They are very complex, and not an economical theory of large-scale phenomena.
An aesthetic objection.
- ❷ They have empirical components (so not wholly a theory).
- ❸ They are imperfect.
 - They are not always in quantitative agreement with observations.
 - They appear to fail miserably in some past climates (equable climates).
- ❹ They work by cancellation of errors.

Corollary and Practical Consequence

- They are not always trusted.
- They are hard to improve.

The Crux of the Matter

If models (simulations) were perfect, then conceptual theory and understanding would be an indulgence, a mere intellectual conceit, of little value beyond personal satisfaction.